EXCHANGE RATE MODELLING

I. INTRODUCTION:

The liberalization of global capital markets, coupled with the sharp rise in exchange rate volatility after the collapse of the Bretton Woods system, created an urgent need for reliable tools to manage foreign exchange risk. As noted by Chalupa (1982), the transition from fixed to floating exchange regimes introduced considerable uncertainty into international trade and investment, thereby accelerating the evolution of derivative products such as currency futures. First introduced at the International Monetary Market (IMM) of the Chicago Mercantile Exchange in the early 1970s, currency futures offered a standardized, transparent, and exchange-traded alternative to traditional over-the-counter forward contracts. With the growth of cross-border trade, capital mobility, and electronic trading infrastructure, these instruments further developed into currency index futures, allowing investors to hedge or speculate on the performance of multiple currency pairs through a single composite contract.

The study of currency futures indices is particularly relevant due to their ability to consolidate movements across multiple currency pairs into a single tradable instrument. Unlike bilateral exchange rates, which reflect isolated currency relationships, index-based futures offer a broader perspective of a currency's strength or weakness against a diversified basket. This broader framework makes them more effective tools for managing exposure, as they enhance hedging efficiency and reduce the influence of idiosyncratic or country-specific risks. Moreover, their composite nature supports systematic trading approaches and macro-level hedging strategies that align with global market movements. Empirical evidence also shows that currency index futures tend to exhibit distinctive patterns of volatility clustering and co-movement with other major asset classes, highlighting their growing relevance in risk modelling, portfolio diversification, and market forecasting.

Recent developments in global financial markets further underscore this importance. Global foreign exchange (FX) trading volumes surged to an unprecedented US$9.6 trillion per day as of April 2025, representing a 28% increase from April 2022. In parallel, the Indian foreign exchange market was valued at US$30.7 billion in 2024, with projections by IMARC Group indicating growth to US$65.8 billion by 2033 at a CAGR of 8.8% between 2025 and 2033. These figures reflect increasing reliance on currency instruments, both for transaction-related hedging and speculative positioning.

The value of currency index futures lies in their ability to encapsulate market expectations, risk premia, and volatility across multiple currencies. Cornett, Schwarz, and Szakmary (1995)[2] classify foreign exchange futures among the most liquid financial instruments worldwide—often surpassing daily volumes of equity and commodity derivatives. Their intraday analyses reveal systematic trading behaviours influenced by macroeconomic news releases, time-zone overlaps, and institutional activity. These patterns reinforce the role of currency futures not only as hedging tools but also as channels of information transmission within the global financial system.

Research on currency futures has evolved significantly over the past four decades, reflecting both the deepening of global financial markets and advances in econometric modeling. Early empirical works focused on assessing the hedging effectiveness and volatility characteristics of currency futures in developed economies. Dale (1981) demonstrated that these instruments substantially reduced exposure to exchange rate risk in the United Kingdom, Germany, and Japan, confirming their value for portfolio risk management. Hodrick and Srivastava (1985) expanded upon this by showing that futures and forward prices are broadly equivalent, apart from minor deviations that capture speculative expectations. Similarly, Harvey and Huang (1991) examined volatility across U.S., European, and Japanese currency futures and observed that fluctuations were higher during U.S. trading hours, indicating that market activity

and information flow significantly influence volatility patterns. Collectively, these studies established the empirical and theoretical foundations of currency futures as efficient tools for hedging and as mechanisms for price discovery in mature markets.

In the 2000s, the literature shifted toward exploring dynamic relationships between spot and futures prices using advanced econometric models. Chakraborty and Barkoulas (2001) evaluated dynamic hedging models and concluded that while these frameworks enhance statistical precision, their real-world performance is constrained by transaction costs and market volatility. Lien and Yang (2006) identified asymmetric effects in spot–futures correlations, showing that the relationships between these markets evolve over time. Rosenberg and Traub (2006, 2008) emphasized that futures markets tend to lead spot prices in the price discovery process, reinforcing their informational efficiency. Meanwhile, Rothig (2004)[9] examined the behavior of currency futures during crises and found that trading activity often intensifies before episodes of financial instability, suggesting that futures markets can act both as early warning indicators and amplifiers of volatility.

Following the introduction of exchange-traded currency futures in India in 2008, research attention turned toward the dynamics of emerging markets. Kaur, Chahal, and Kharb (2014) found weak-form inefficiency in USD/INR futures, indicating that the market was still developing and adjusting to new instruments. Kumar (2015) reported that the launch of currency futures initially increased volatility but later contributed to greater stability as market participants gained experience and liquidity improved. Subsequently, Kharbanda and Singh (2018) employed CCC-MGARCH models to examine efficiency and hedging effectiveness, concluding that Indian currency futures markets have become increasingly efficient and effective, particularly for USD/INR contracts.

The motivation for this study arises from the growing significance of currency futures as integral instruments in global financial risk management. In the context of increasing capital mobility, exchange rate volatility, and cross-border financial integration, the ability to hedge foreign exchange exposure efficiently has become a fundamental requirement for firms and institutional investors alike. Currency futures, by virtue of their standardized contracts, exchange-traded transparency, and price discovery capabilities, provide a vital mechanism for mitigating exchange rate uncertainty and enhancing market efficiency. The evolution of currency index futures has further expanded the analytical scope, enabling the examination of aggregate currency movements and their interdependencies with other financial markets. In emerging economies such as India, where derivative markets are still in a maturing phase, studying the performance, efficiency, and volatility implications of currency futures assumes particular importance. This research is therefore motivated by the need to understand how these instruments contribute to financial stability, risk transmission, and policy effectiveness in a rapidly evolving global monetary environment.

II. EVOLUTION OF CURRENCY FUTURES:

1. Introduction

In today's global economy, currencies constantly change value due to trade flows, interest rates, economic policies, and geopolitical events. For businesses, investors, and governments that deal internationally, this volatility can impose major risks. Currency futures were developed as financial tools to reduce these risks by allowing parties to lock in exchange rates ahead of time. Over decades, these instruments have evolved from nascent beginnings following fixed−currency regimes to robust, globally−traded products with advanced technology, regulation, and innovation.

2. Historical Background

Before Floating Rates and the Need for Futures

For much of the 19th and early 20th centuries, many countries operated under somewhat fixed or pegged currency systems (e.g. gold standard, and later Bretton Woods). Under such regimes, currency values were relatively stable or adjusted only by formal policy decisions. Because of that, while there was some currency risk, it was limited and predictable. The major risks for exporters/importers or investors came from tariffs, wars, inflation, etc., but sudden large shifts in exchange rates were less common.

The Shift: Floating Rates and Problems

In the early 1970s, especially after the collapse of Bretton Woods (1971−73), many major currencies began "floating" against each other. This meant their exchange rates were no longer fixed by governments but determined by market forces. While this allowed greater flexibility for monetary policy, it also introduced much more uncertainty:

Businesses that imported or exported goods suddenly faced the risk that when payment is due months later, the exchange rate could have changed badly for them.

Investors holding assets denominated in foreign currencies could face losses simply due to adverse currency movements.

Hedging via forward contracts (over−the−counter, bespoke deals) was possible but had drawbacks: counterparty risk, low liquidity, limited accessibility for smaller players, and lack of standardization.

Birth of Currency Futures

To deal with these problems, the financial markets introduced currency futures. In 1972, the Chicago Mercantile Exchange (CME) launched the first standardized currency futures contracts. These allowed contracting today to buy or sell a fixed amount of a currency at a fixed exchange rate at a set future date. Key features included:

Standardized contract sizes, delivery/maturity dates.

Being traded on an exchange which reduced counterparty risk via a clearing house.

Transparent pricing and more liquidity, so participants could more easily enter or exit positions and have better knowledge of prevailing rates.

Thus, currency futures became the "fix" to many of the uncertainties introduced by floating rates.

3. Today vs Earlier Times: How Things are Different and Problems Are Solved

Since their introduction, currency futures have solved several earlier pain points.

- **Hedging becomes practical and safer:** Businesses can lock in exchange rates, reducing the risk of unpredictable currency swings.
- **Reduced counterparty risk:** Because exchanges and clearinghouses stand in between buyer and seller, there's less risk that one party fails (compared to bespoke forward contracts).
- **Greater transparency and price discovery:** Market prices are publicly visible, enabling all participants to see what rates the market is valuing.
- **Liquidity and standardization:** Major currency futures (EUR/USD, USD/JPY, etc.) are heavily traded; because contracts are standardized, it's easier to trade them.
- **Accessibility**: Over time, more currencies have been added; technological platforms have made trading easier for non−large institutional players.

How Different It Is from Earlier Times

- **Fixed vs Floating Regimes:** Earlier systems tried to fix or peg currencies; now floating is standard, so currency risk is unavoidable, making tools like futures essential.
- **OTC vs Exchange−Traded:** Before, forward FX contracts were negotiated bilaterally. Now, many hedging transactions are done via exchange−traded currency futures.
- **Manual vs Electronic Trading**: In earlier times, much of financial trade was manual and slower. Now, trading is digital, often automated or algorithmic, with near−real−time data and global participation.
- **Limited Currency Choices vs Wide Universe:** Originally few major currency pairs had futures contracts. Now many emerging market currencies and more complex pairings are available.
- **Information & Regulation:** Earlier, market participants might have limited data and little regulatory oversight. Today, there is more regulation, more oversight, reporting, margin requirements, etc., which bring both stability and cost.

III. REVIEW OF LITERATURE:

Currency futures have become essential tools for managing exchange rate risks and improving the overall efficiency of financial markets. Since their emergence in the 1970s, these instruments have attracted significant attention from researchers and have been the subject of extensive empirical and theoretical research exploring how currency futures contribute to price discovery, influence volatility, enhance hedging strategies, and strengthen market efficiency. The growing focus of the researchers globally as well as in the Indian context reflects the increasing importance of currency futures in shaping financial stability and investor behavior in diverse market settings.

The review of literature draws together insights from some of the most influential academic studies on currency futures, exploring how different researchers have approached the topic and how key themes have developed over time. By bringing together global perspectives alongside research from India, it shows how understanding of currency futures has steadily evolved from theoretical discussions to more data-driven, real-world analyses.

GLOBAL PERSPECTIVE

Hodrick and Srivastava (1985) were among the first researchers to take a deep and systematic look at how foreign currency futures actually behave in comparison to forward contracts. Their study combined both theory and data to explore whether these two instruments, which are used for managing exchange rate risk, generally move together. They found that in most cases, futures and forward prices tend to align quite closely, lending support to the efficient market hypothesis (EMH), which assumes that all available information is already reflected in market prices. However, they also noted that small discrepancies can arise, often due to practical factors like transaction costs, differences in contract settlement mechanisms, and variations in traders' expectations about future exchange rate movements.

In a similar vein, Dale (1981) explored the usefulness of currency futures as hedging tools by examining markets in the UK, Germany, and Japan. His research focused on whether these instruments could genuinely reduce the risks associated with unpredictable exchange rate movements. His findings showed that currency futures offered solid protection against exchange rate swings, especially in markets with frequent or sharp fluctuations. He also found that the success of hedging differed across countries, depending on how liquid and developed each market was. This made his work especially useful for global investors and firms looking to manage currency risk more effectively.

Furthermore, Harvey and Huang (1991) examined the volatility behavior in major foreign exchange futures markets (U.S., European, Japanese) and discovered that volatility tends to rise during U.S. trading hours, a period marked by high trading activity and a rapid flow of new economic information, reflecting liquidity and information asymmetry dynamics. Their quantitative empirical approach set a precedent for later econometric analyses focusing on volatility clustering and risk transmission.

The study of currency futures became far more sophisticated with the works of Chakraborty and Barkoulas (2001) and Lien and Yang (2006), who introduced advanced econometric techniques to better understand how these markets function. Instead of assuming that market relationships remain stable over time, they used dynamic hedging models and time-varying correlation frameworks to reflect the reality that financial markets constantly evolve. Dale's analysis emphasizes that hedgers can select futures contracts with varying maturities and adjust their exposures as payment dates approach or as risk estimates shift, which is a key aspect of dynamic hedging. The core findings of Dale (1981) demonstrate that currency futures markets, specifically for British pounds, German marks, and Japanese yen, are highly effective hedging instruments, thus, achieving risk reduction rates exceeding 97% in both two-week and four-week hedging intervals. This level of effectiveness surpasses or matches the hedging performance of long-established agricultural futures markets, such as wheat and corn.

Building on this, Lien and Yang (2006) took the analysis a step further by uncovering asymmetries in the spot–futures relationship. They found that markets react differently to positive and negative shocks. This meant that gains and losses do not affect hedge ratios or risk exposure in the same way. This insight added a more nuanced understanding of how traders and firms should adapt their strategies under varying market conditions. Together, these studies marked a shift toward a more realistic and flexible approach to modeling and managing currency futures, aligning econometric research more closely with actual market behavior.

Rothig (2004) extended the discussion to crisis contexts, analyzing multiple countries (Australia, Canada, Japan, Korea, Switzerland) and demonstrating that futures trading activity tends to amplify spot market volatility during currency crises, revealing potential destabilizing effects when speculative trading dominates. This observation challenged the conventional assumption that futures markets uniformly enhance market stability. However, in loosely regulated or sentiment-driven environments, the same mechanisms can amplify systemic risk.

Furthermore, Rosenberg and Traub (2006, 2008) added a new layer to the understanding of currency futures by focusing on the market microstructure, or the way information and trading activity shape price movements. They positioned currency futures as active arenas of information generation and dissemination, where market expectations are continuously shaped and transmitted. They discovered that futures prices often move first, especially during periods of intense trading, suggesting that futures. Their research not only advanced the discourse on price discovery but also underscored the critical role of transparent, liquid futures markets in enhancing overall market efficiency and reducing informational delays.

INDIAN PERSPECTIVE

The introduction of currency futures in India marked a major reform in the country's financial architecture.

Deepika Upadhyay (2009) provides one of the first analytical looks at India's currency futures market. She highlights the launch of USD/INR exchange-traded contracts on 29 August 2008 and the key roles of the RBI and SEBI in setting up regulatory norms. The contracts were standardized, offering corporates and individuals a transparent hedging tool. Upadhyay notes that India's cautious approach, including limits on foreign investors, helped maintain stability in the early stages. She also emphasizes that expanding participation and introducing more currency pairs would be important for improving long-term liquidity.

A more microeconomic approach to understanding exchange-rate dynamics in India was undertaken by N. R. Bhanumurthy (2004) which bridges the gap between macroeconomic fundamentals and market microstructure theory by examining how dealer behavior, order flow, and bid-ask spreads affect short-term movements in the INR/USD exchange rate. Using survey data from forex dealers and empirical time-series analysis, Bhanumurthy concludes that microstructure variables such as the number of trades and trading volume explain short-run exchange-rate changes far better than macroeconomic indicators like interest rate differentials or inflation. He also observes that speculative trading, contrary to common perception, can improve market efficiency, whereas excessive central bank intervention may distort price discovery.

Further, as currency futures gained momentum post-2008, researchers began exploring their effect on spot market volatility. Dhananjay Sahu (2012) examined whether the introduction of exchange-traded derivatives influenced volatility in the EUR/INR market between January 2008 and December 2011 employing econometric techniques such as ARCH-LM and GJR-GARCH (1,1) models. His results emphasize that the effectiveness of derivatives in reducing volatility depends on market depth and the diversity of participants.

Further exploring price discovery, Amandeep Kaur and Narinder Pal Singh (2024) used daily EUR/INR spot and futures data from 2010 to 2019, applied the Augmented Dickey-Fuller (ADF) unit root test, Johansen cointegration

analysis, Granger causality, and Vector Error Correction Models (VECM) and found a long-run equilibrium relationship between futures and spot markets, with Granger causality running from futures to spot, indicating that futures markets lead spot rates in terms of information content. The study concludes that exchange-traded currency futures contribute significantly to price discovery and overall market efficiency in India.

Complementing these studies is work like Amit Mirchandani (2013), whose paper connects macroeconomic fundamentals such as inflation, trade balance, and interest-rate differentials with INR volatility. His analysis suggests that while structural macro variables dominate long-term trends, market-based instruments like futures primarily influence short-term volatility and expectations. This complements the findings of Bhanumurthy and Sahu by distinguishing between macroeconomic and microstructure influences.

Kaur, Chahal, and Kharb (2013) employed statistical tests such as the Runs Test and Ljung–Box Q-statistics to assess weak-form efficiency in USD/INR futures. Their results suggested that the market did not fully exhibit efficiency in the initial years, implying potential opportunities for arbitrage and improvement in market depth[12]. Similarly, Kumar (2015) analyzed the volatility impact of currency futures on the Euro/INR exchange rate, identifying a reduction in volatility post-introduction of futures contracts, indicating their stabilizing influence on the spot market.

Furthering this line of inquiry, Kharbanda and Singh (2018) applied CCC-MGARCH modeling to evaluate both efficiency and hedging effectiveness. Their findings revealed that advanced multivariate GARCH models outperform traditional OLS-based hedge ratios in minimizing portfolio risk, aligning Indian evidence with global econometric advancements.

Collectively, these studies present a nuanced picture of the Indian currency futures landscape. The literature reveals that the 2008 introduction of exchange-traded currency derivatives transformed India's foreign exchange ecosystem by improving transparency and facilitating price discovery. However, the degree of effectiveness depends on the specific currency pair, market liquidity, and policy environment. While early works like Upadhyay (2009) emphasized regulatory prudence and market development, later econometric analyses focused on the market's quantitative impacts—such as volatility, causality, and efficiency. Meanwhile, Bhanumurthy (2004) and Mirchandani (2013) provided deeper theoretical and structural insights, underscoring how microstructure and macroeconomic forces jointly determine exchange-rate dynamics in India.

Thus, the review of literature consistently highlights that currency futures in India play a crucial role in helping market participants manage risk and discover prices. They provide a structured mechanism for hedging against exchange rate fluctuations, which is particularly valuable for businesses and investors exposed to currency movements. However, their ability to stabilize market volatility is not uniform; it depends on factors such as the prevailing market conditions, the diversity of participants, and the depth of liquidity.

IV. THEORETICAL UNDERPINING:

Currency futures, often called foreign exchange (FX) futures, are standardized contracts that allow one to buy or sell a certain amount of one currency against another at a fixed price on a future date.
The idea behind these instruments comes from a mix of international finance theory, pricing models for derivatives, and the concept of efficient markets.
These theories together help us understand how the prices of currency futures are decided, how arbitrage keeps markets balanced, and how traders and investors use these contracts for hedging, speculation, and risk management.

2. Interest Rate Parity (IRP) Theory

The Interest Rate Parity (IRP) theory is one of the most important ideas behind how currency futures are priced. It basically says that the difference between the spot exchange rate (the current rate) and the futures or forward exchange rate depends on the difference between the interest rates of the two countries involved.
If this relationship didn't hold, arbitrageurs could make easy, risk-free profits by borrowing in one currency and investing in another which in real markets quickly gets corrected.

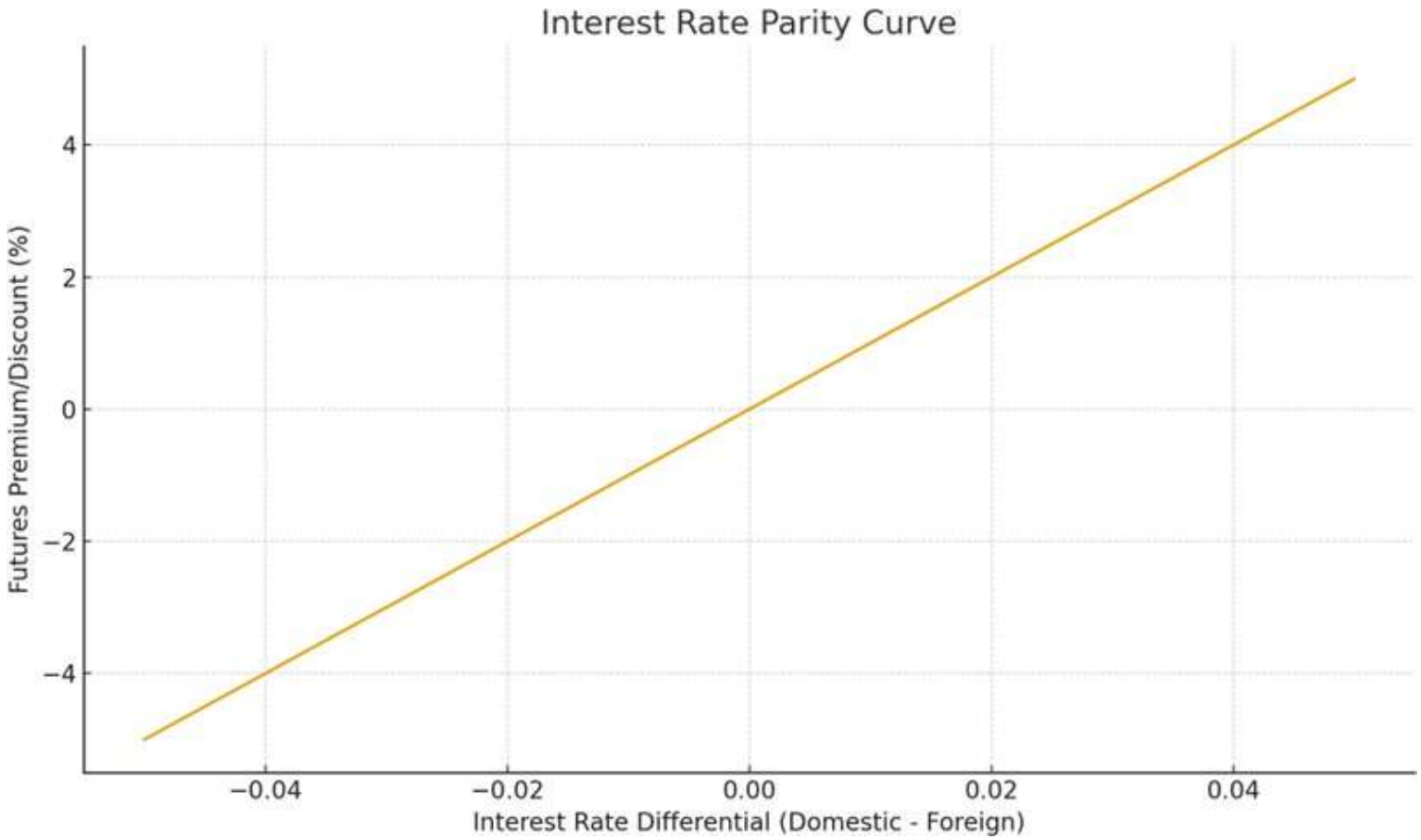
Formula:

$F=S\times(1+i_d) / (1+i_f)$

Here,

- S = spot exchange rate
- F = futures exchange rate
- i_d = domestic interest rate
- i_f = foreign interest rate

In simple words, if the domestic country has higher interest rates, its currency will generally trade at a discount in the futures market. If its rates are lower, it will trade at a premium.
A typical IRP graph would show that as the domestic interest rate rises above the foreign one, the domestic currency's futures price moves to a discount keeping the market in equilibrium.



The Interest Rate Parity (IRP) curve, showing the relationship between the interest rate differential (domestic minus foreign) and the futures premium or discount. The graph shows that as the domestic interest rate exceeds the foreign rate, the domestic currency futures trade at a discount, and vice versa. This graph shows how the IRP works, there are no easy profits from differences between spot and futures prices.

3. Cost of Carry Model

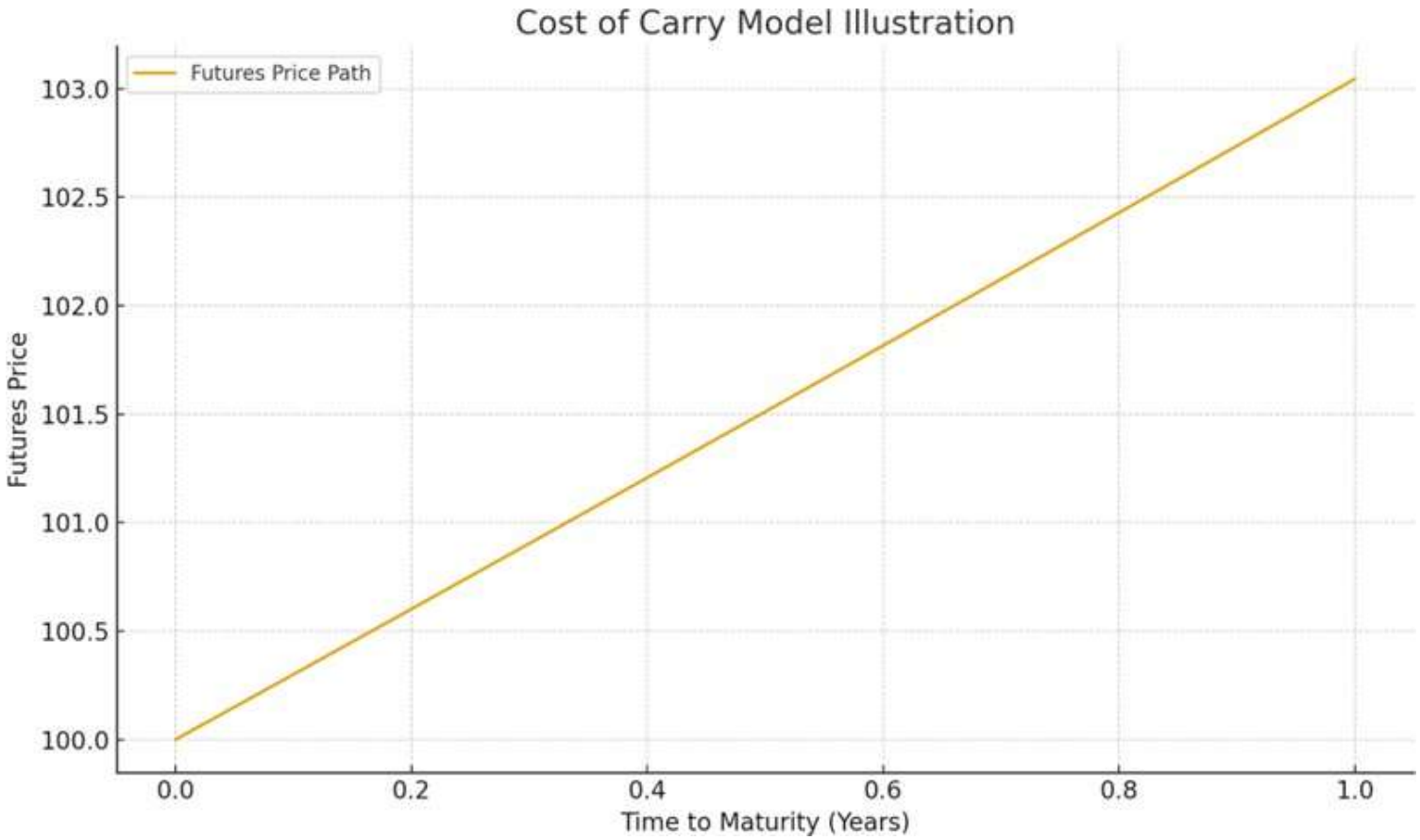
The Cost of Carry Model is another key concept that links the spot price of a currency to its futures price. It is based on the idea of the "cost" or "benefit" of holding one currency instead of another.
In formula form (under continuous compounding):

$F=S\times e^{((r_d-r_f)T)}$

where,

- r_d = domestic interest rate
- r_f = foreign interest rate
- T = time left until the contract matures

If domestic interest rates are higher, the futures price will usually be higher than the spot price, because holding domestic currency has an opportunity cost.
So as time passes, the futures price tends to rise compared to the spot price, showing how interest rate differences influence currency values.



The Cost of Carry model, showing how the futures price of a currency evolves over time relative to the spot price. The graph starts at the spot price (100) and gradually rises, reflecting the positive interest rate differential between the domestic and foreign currencies (domestic rate = 6%, foreign rate = 3%). As the time to maturity increases, the futures price increases due to the opportunity cost or benefit of holding one currency versus another.

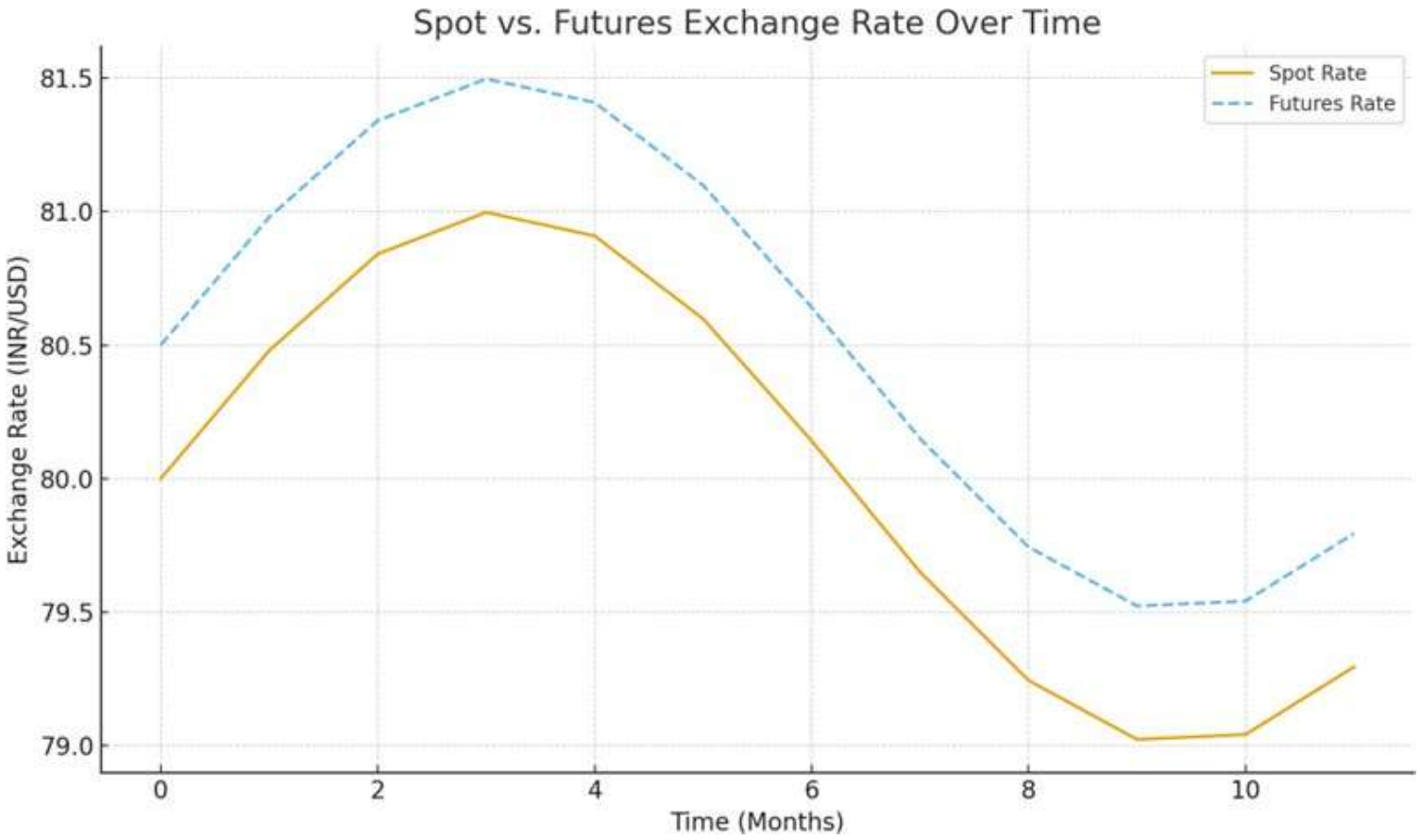
4. Expectations Hypothesis

The Expectations Hypothesis says that the futures rate reflects what the market expects the spot exchange rate to be in the future.
In other words, the futures price is seen as an unbiased forecast of the future spot rate — assuming everyone in the market acts rationally and there's no extra risk premium involved.

Formula:

$E(S_{t+T})=F_t$

However, in real life, things are rarely perfect. Because traders often prefer safety over risk, the futures price can differ slightly from the actual future spot rate. That small gap represents a risk premium — the reward for taking on uncertainty.



This graph compares the spot exchange rate and the corresponding futures rate over a 12-month period. The futures price generally tracks the spot rate, with a small premium reflecting the interest rate differential between the two currencies. This graph shows the Expectations Hypothesis, showing that futures prices act as an unbiased predictor of the expected future spot rate, assuming rational market behavior.

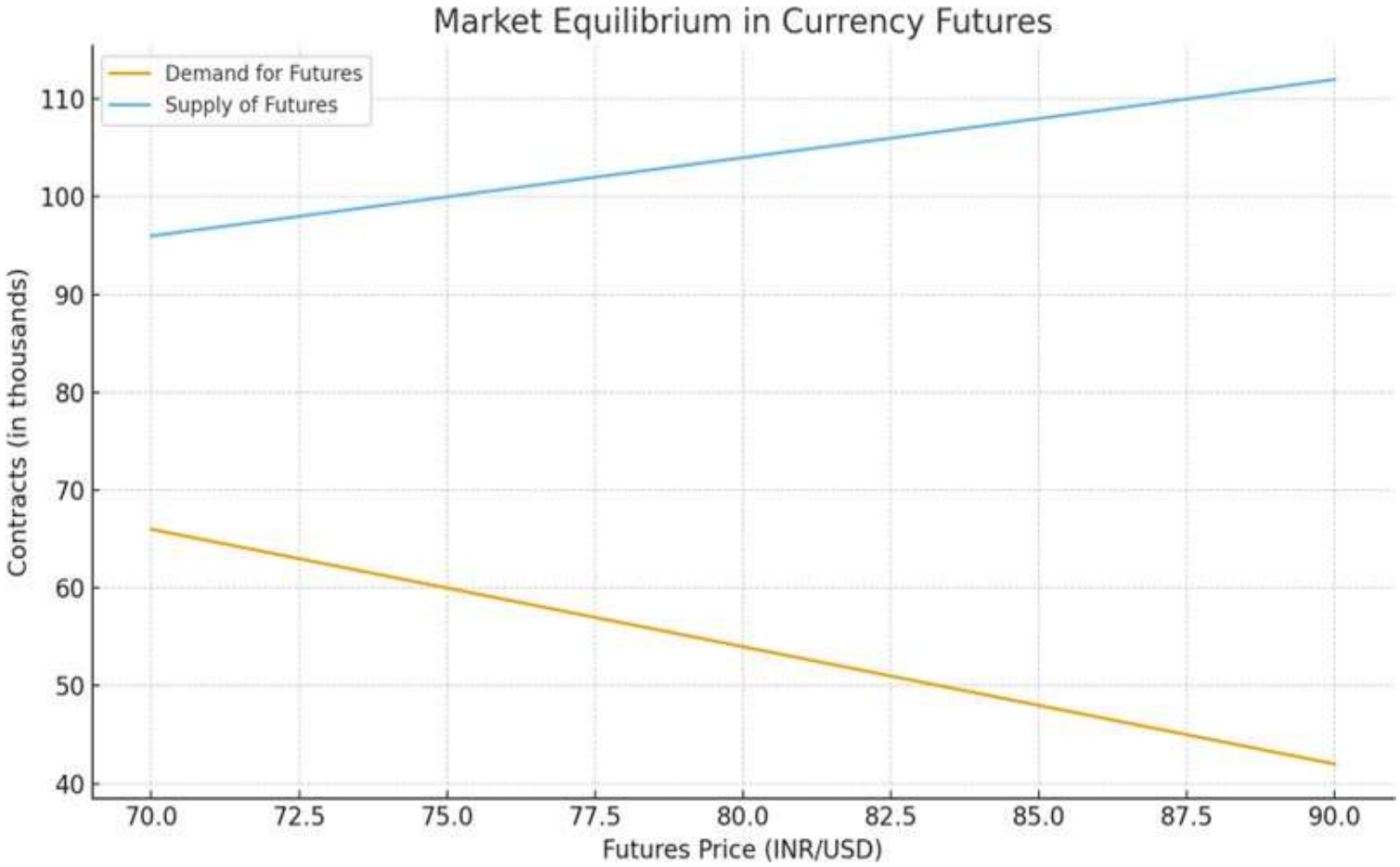
5. Efficient Market Hypothesis (EMH)

The Efficient Market Hypothesis (EMH) argues that all available information about exchange rates is already built into the current prices of currency futures.
This means prices adjust almost instantly when new information comes out, making it nearly impossible to consistently predict or profit from future price movements.
In short, the EMH supports the idea that currency futures markets are fair and efficient, and that any profit opportunities disappear quickly once traders act on them.

6. Arbitrage Pricing and Hedging Theory

Arbitrage plays a key role in keeping the relationship between the spot and futures exchange rates stable.
Whenever prices move out of line with what the theory suggests, arbitrageurs step in to take advantage of the difference , buying in one market and selling in another , until prices return to balance.
On the other hand, hedgers use currency futures to protect themselves from unfavorable exchange rate changes. Their goal isn’t to make a profit but to reduce uncertainty.

Both arbitrage and hedging are guided by ideas from portfolio theory, which focuses on finding the right balance between risk and return.



The equilibrium in the currency futures market through the interaction of demand and supply for futures contracts. The downward-sloping demand curve shows that buyers purchase fewer contracts as the futures price rises, while the upward-sloping supply curve shows that sellers are willing to provide more contracts at higher prices.

7. Supporting Theories in Currency Futures Pricing

Apart from the main theories, a few other concepts also help explain how currency futures are priced:

- Purchasing Power Parity (PPP): Says that exchange rates adjust over time to reflect inflation differences between countries.
- Covered Interest Arbitrage (CIA): Shows how IRP holds true in real markets through actual trading activities.
- Uncovered Interest Parity (UIP): Focuses on how interest rate differences relate to expected future spot rates.
- Rational Expectations Theory: Suggests that market participants use all available information efficiently when forming expectations.
- Capital Market Theory: Explains how futures contracts help in diversifying risk and improving portfolio performance.

8. Practical Implications of Currency Futures Theory

All these theories have very practical uses in real-world markets:

- Hedgers use currency futures to lock in exchange rates and avoid losses from currency fluctuations.
- Speculators use them to try to profit from expected movements in exchange rates or interest rates.
- Arbitrageurs look for short-term mispricings between the spot and futures markets and correct them, keeping prices aligned.

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V. MARKET ANALYSIS:

→ Market Size and Volume:

Source of data:

1. Definitions:

Before analysing the specific figures, it is crucial to establish precise definitions for the terms used in the context of a currency futures market. The terms "size" and "volume" can be ambiguous and are often used interchangeably. Based on the data provided, we can deduce a clear and logical interpretation.

- Currency Futures: A currency futures contract is a legally binding agreement to buy or sell a specific amount of a foreign currency at a predetermined exchange rate (the "futures price") on a specific date in the future (the "settlement date"). These instruments are used by corporations for hedging against exchange rate risk and by traders for speculation.
- Market Volume (Turnover in USD): In the context of our data, this figure represents the total notional value, denominated in US Dollars (USD), of all contracts traded during the period.
    - *Example:* Our figure of 194,803,950 for 2024 likely represents $194.80 million in total notional value traded.
- Market Size (Turnover in INR): This figure represents the total notional value, denominated in Indian Rupees (INR), of all contracts traded. It is the INR equivalent of the "Market Volume."
    - *Example:* Our figure of ₹16,26,85,69,038.40 for 2024 represents ₹16.27 billion in total turnover.
- Average Market Size (Average Daily Turnover): This is arguably the most important metric for comparative analysis. It is calculated by dividing the Total Market Size (Turnover) by the number of trading days in the period. It provides a "like-for-like" comparison of market activity, neutralizing the effect of comparing a full year (2024) to a partial year (2025).

Important Distinction: Turnover vs. Volume (Number of Contracts)

It is important to note that our data appears to represent Turnover (monetary value), not Volume (number of contracts).

- Volume (Contracts): This is a simple count of how many contracts were traded (e.g., 500 contracts).
- Turnover (Value): This is the total monetary value of those contracts (e.g., 500 contracts * $1,000/contract * ₹87/USD = ₹43,500,000).

Our data, where one figure is in USD and the other is its INR equivalent, clearly represents Turnover.

2. Interpretation of Data

Based on the definitions above, we can interpret our calculations to find a third, highly valuable metric: the Implied Average Exchange Rate. This is found by dividing the INR turnover by the USD turnover.

2.1 Data for Full-Year 2024

- Total Market Volume (Turnover in USD): $194,803,950.00
- Total Market Size (Turnover in INR): ₹16,26,85,69,038.40
- Average Market Size (Avg. Daily Turnover, INR): ₹6,75,04,435.84
- Implied Average Exchange Rate (2024):
    - Total Size (INR) / Total Volume (USD)
    - ₹16,26,85,69,038.40 / $194,803,950.00 = ₹83.51 per USD

This implied rate of ₹83.51 represents the weighted average exchange rate at which futures contracts were traded throughout 2024.

2.2 Data for 2025 (YTD to March 6)

- Total Market Volume (Turnover in USD): $22,644,500.00
- Total Market Size (Turnover in INR): ₹1,96,49,70,489.50
- Average Market Size (Avg. Daily Turnover, INR): ₹4,36,66,010.88
- Implied Average Exchange Rate (2025):
    - Total Size (INR) / Total Volume (USD)
    - ₹1,96,49,70,489.50 / $22,644,500.00 = ₹86.78 per USD

This implied rate of ₹86.78 represents the weighted average exchange rate for trades in the first 45-46 trading days of 2025.

3. Critical Analysis of Market Trends (2024 vs. 2025)

By comparing the 2024 data with the 2025 YTD data, two significant trends emerge: a sharp decrease in market activity and a clear depreciation of the INR.

3.1 Analysis of Market Activity (Liquidity)

The most effective comparison is between the average daily figures, as this normalizes the different time periods.

- Average Daily Turnover (INR) 2024: ₹6.75 crore
- Average Daily Turnover (INR) 2025 YTD: ₹4.37 crore

Observation: There has been a 35.2% decrease in the average daily turnover in the INR futures market. ([₹4.37cr - ₹6.75cr] / ₹6.75cr = -0.352)

Interpretation: This is a substantial contraction in market activity. It indicates that significantly less value is being traded on a daily basis in early 2025 than in 2024. This suggests:

- Lower Liquidity: With less money flowing, it may be harder to enter or exit large positions without affecting the price.
- Reduced Market Participation: This could imply that both hedgers (importers/exporters) and speculators are less active.

- Possible "Wait-and-See" Approach: Market participants might be holding back due to global economic uncertainty, waiting for clearer signals from central banks (like the RBI or the U.S. Federal Reserve), or anticipating future volatility.

3.2 Analysis of Currency Valuation

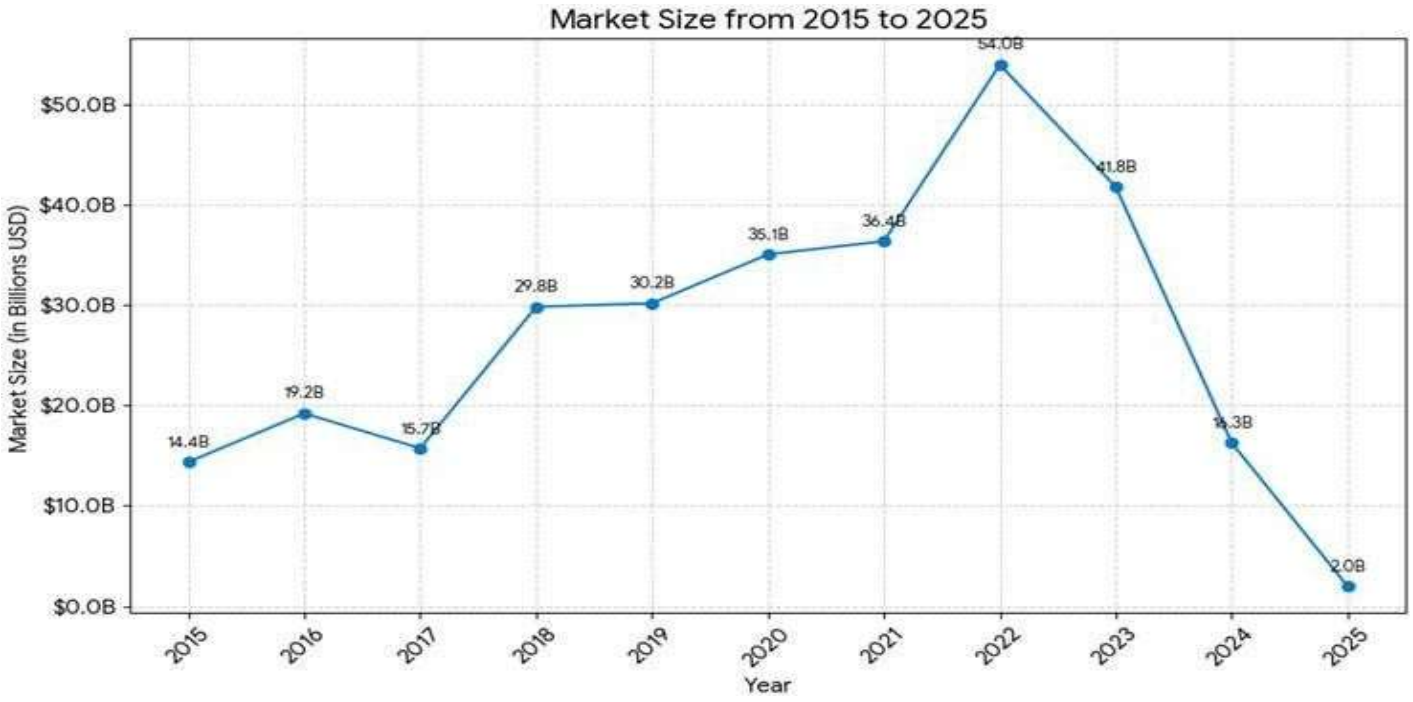
The implied exchange rates calculated from our data provide a clear picture of the currency's directional trend.

- Implied Average Exchange Rate 2024: ₹83.51 / USD
- Implied Average Exchange Rate 2025 YTD: ₹86.78 / USD

Observation: The USD has become 3.9% more expensive in INR terms in early 2025 compared to the 2024 average. ([₹86.78 - ₹83.51] / ₹83.51 = +0.039)

Interpretation: This confirms a clear trend of INR depreciation (or USD appreciation). This is consistent with the price data from the USD_INR_Combined_Prices (2).csv file you previously mentioned, which shows futures prices in the 86-87 range. This trend is driven by macroeconomic factors such as interest rate differentials, capital flows, inflation, and trade balances.

4. Analysis and Interpretation of Market Size and Volume (2015-2025)

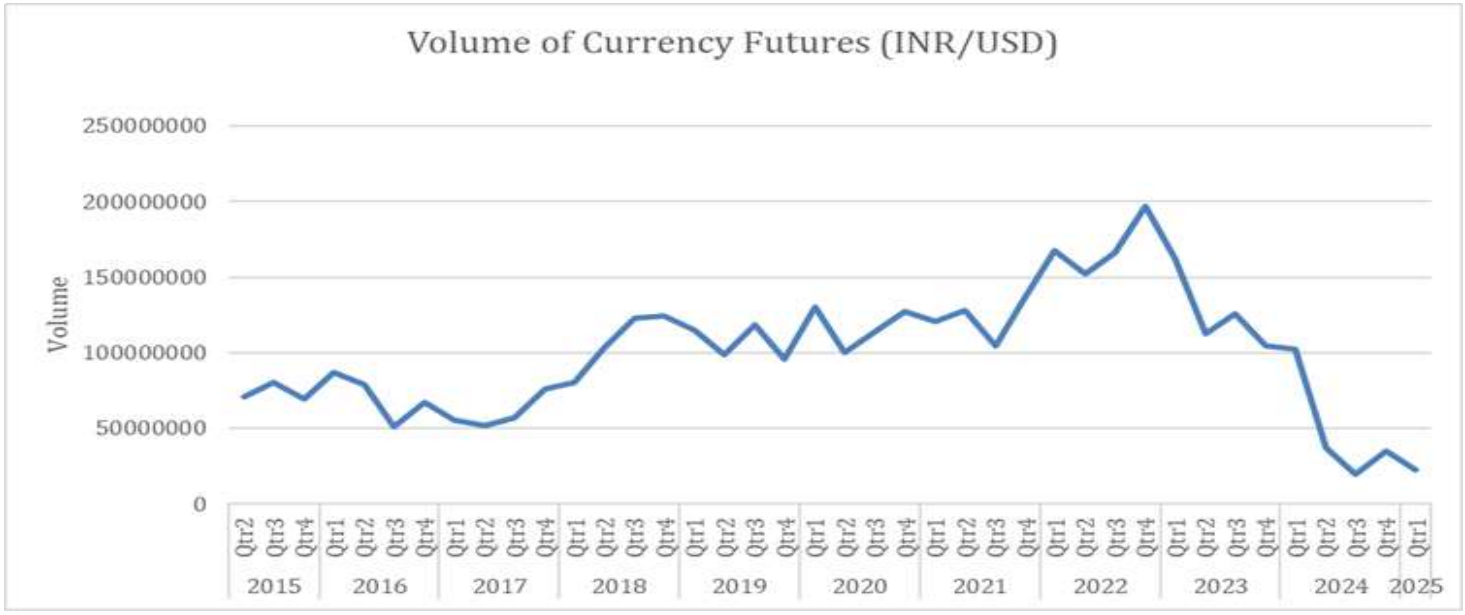


Graph 1: Analysis of Market Volume (Turnover in USD)

This graph illustrates the total notional value of all contracts traded, denominated in US Dollars. This is the most accurate measure of pure market activity and liquidity, as it is not affected by changes in the USD/INR exchange rate.

Key Observations:

1. Cyclical Activity: The market has not grown linearly. It exhibits significant cyclicality, with activity peaking in 2018-2019, followed by a sharp contraction in 2021, and then a strong recovery to a new peak in 2024.
2. Peak and Troughs:
    - Peak Activity: 2024 saw the highest trading volume in this 10-year period, with $194.80 million in turnover. This surpasses the previous peaks of 2019 ($190.92M) and 2018 ($183.17M).
    - Significant Lows: The lowest activity was in 2015 ($114.36M). Notable contractions occurred in 2017 ($128.49M) and 2021 ($133.03M).
3. Recent Trend: After the 2021 low (which may be attributed to economic disruptions from the COVID-19 pandemic), the market saw a robust three-year recovery (2022-2024). This suggests a strong return of both hedgers (importers, exporters) and speculators, likely driven by increasing global economic uncertainty and currency volatility.



Graph 2: Analysis of Market Size (Turnover in INR)

This graph represents the INR equivalent of the USD turnover. Its value is a function of two variables: the Market Volume (USD) and the prevailing USD/INR exchange rate.

Key Observations:

1. High Correlation: As expected, the visual pattern of this graph is highly correlated with the Market Volume (USD) graph. The peaks (2018-2019, 2024) and troughs (2017, 2021) occur in the same years.
2. Amplified Growth (The Exchange Rate Effect): The most critical insight comes from comparing the *relative* growth of the two metrics.
    - In 2018, the peak USD volume was $183.17M, resulting in an INR turnover of ₹12,39.76 crore.
    - In 2024, the peak USD volume was $194.80M, resulting in an INR turnover of ₹16,26.86 crore.

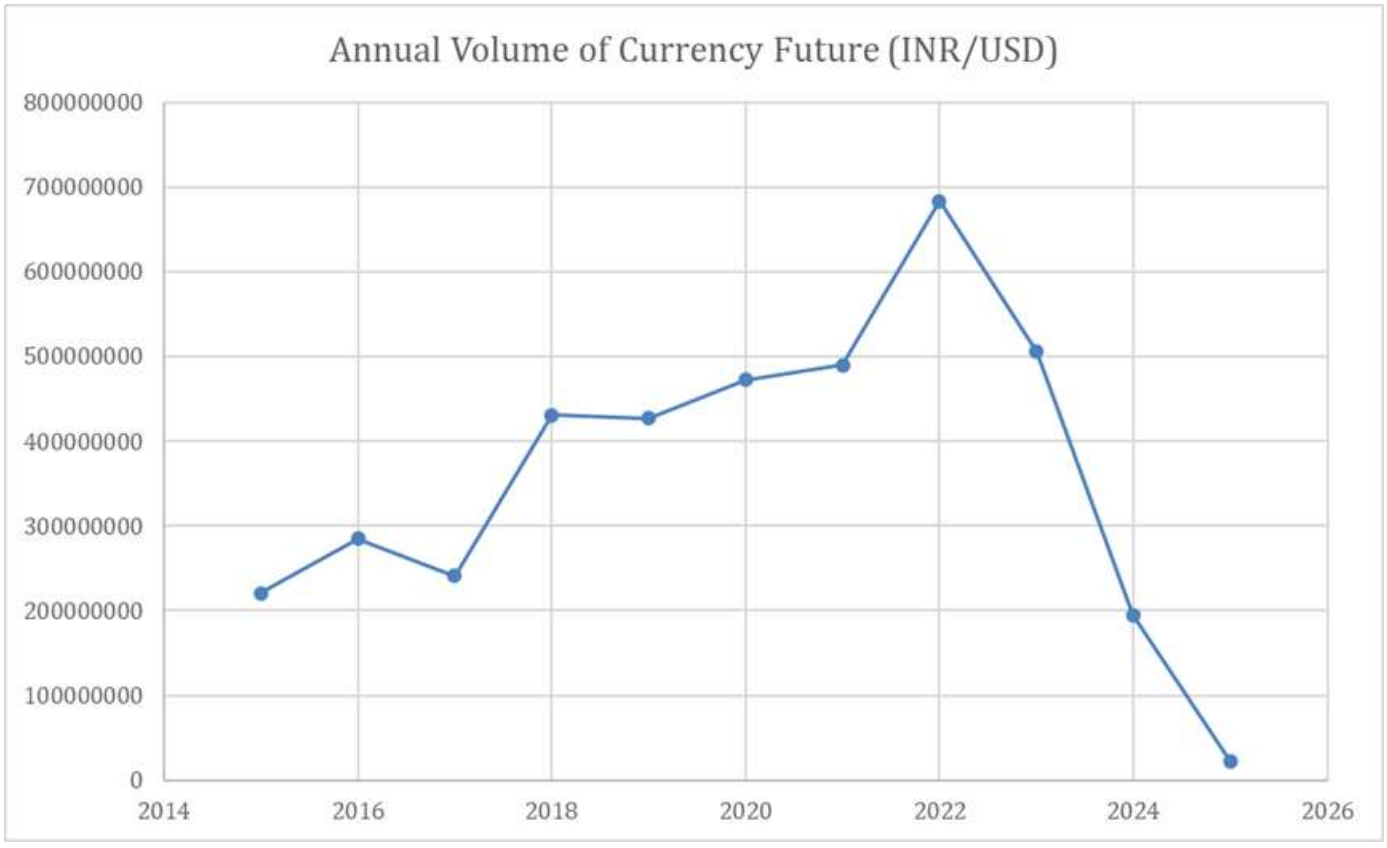
While the USD Volume (activity) grew by only 6.4% between these two peaks (2018 vs. 2024), the INR Size (value) grew by a massive 31.2%.

→ LIQUIDITY ANALYSIS IN THE USD/INR CURRENCY FUTURES MARKET (2015–2025):

This section analyses the annual trading volume of USD/INR currency futures from 2015 to 2025, to understand how liquidity has evolved over time. The annual volume acts as a proxy for market liquidity, since higher trading volume generally indicates tighter bid–ask spreads, easier trade execution, and more efficient price discovery.

1. Overview of the Chart

The Annual Volume chart represents the total contracts traded per year for USD/INR futures. Volume is considered a strong indicator of liquidity, as higher trading activity reflects better market participation and depth. Periods of high trading volume suggest improved liquidity conditions in the market.



2. Year-wise Trend Description3. Liquidity Interpretation

Year	Approx. Volume	Trend / Observation
2015–2017	Low to moderate	Early stage of liquidity buildup gradual increase in participation.
2018–2020	Rising steadily	Market maturing consistent volume growth reflects stronger trader confidence and increased hedging/speculative interest.
2021–2022	Peak years (especially 2022)	Liquidity at its highest; volume surpasses 70 million+ likely due to global macro shocks (post-COVID, inflation, dollar strength, Fed policy changes).
2023	Noticeable decline	Market stabilization reduced volatility leads to lower trading incentives.
2024–2025	Sharp fall (2025 near zero)	Possible data cutoff or actual decline; liquidity dried up fewer participants, reduced uncertainty, or market shift to alternative instruments.

The observed changes in annual volume can be directly linked to market liquidity. The evolution of liquidity can be divided into three broad phases:

- 2015–2020: Liquidity Expansion Phase:

  This phase shows consistent growth in trading activity, reflecting increasing confidence and depth in the USD/INR futures market. Institutional participation rose, hedging demand increased, and market infrastructure improved. Liquidity improved as spreads narrowed, turnover rose, and price discovery became more efficient.

- 2021–2022: Peak Liquidity Phase:

  These years mark the highest liquidity levels. Record trading volumes indicate maximum market depth and participation. High volatility in exchange rates during the post-pandemic period drove strong activity among traders and hedgers. This period highlights a positive link between volatility and liquidity.

- 2023–2025: Liquidity Contraction Phase:

  A significant decline in volume during this phase reflects a fall in liquidity. As volatility reduced, both speculative and hedging activities decreased. Possible causes include stable exchange rates, reduced arbitrage opportunities, policy or margin changes, and a shift toward other derivative products. Liquidity deteriorated, leading to higher transaction costs, wider spreads, and less active trading.

4. Relationship Summary (Volume and Liquidity)

Volume Trend	Liquidity Condition	Interpretation
Rising (2015–2020)	Improving liquidity	Deepening market participation
Peak (2021–2022)	Highly liquid	Active trading and tight spreads
Falling (2023–2025)	Declining liquidity	Calmer market with fewer trades

5. Key Insight

The Annual Volume chart confirms that liquidity in the USD/INR futures market is cyclical and largely driven by volatility. When volatility increases, liquidity tends to rise as traders and hedgers become more active. Conversely, when volatility stabilizes, liquidity declines as trading incentives diminish. The high volatility period between 2019 and 2022 coincided with the highest annual volumes and strongest liquidity conditions.

→ TRADE AND TURNOVER:

The data indicate substantial activity in exchange-traded currency futures across various global financial centres, reflecting the need for currency risk management (hedging) and speculative interest. The US market, dominated by the CME Group, shows vastly higher turnover in notional terms ($101 billion ADNV for FX futures & options) than the other reported figures, highlighting its central role in global FX derivatives trading. Countries like India (NSE), Brazil (B3), and Russia (MOEX) demonstrate considerable activity, either in contract volume or estimated turnover value.

This suggests active participation by domestic and international players managing exposure to currencies like the INR, BRL, and RUB.

Key Market Highlights:

1. United States (CME Group): Clearly the leader in exchange-traded FX derivatives. The high ADNV ($101bn) and ADV (1.1m contracts) underscore deep liquidity, particularly in major currency pairs involving the US dollar. The data combines futures and options.
2. India (NSE): Shows consistent daily activity, primarily focused on the USD/INR pair. While the daily turnover figure (~$27 million) seems small compared to the US, it represents a significant domestic market for hedging Rupee exposure. Monthly figures would provide a better scale comparison.
3. Brazil (B3): Reports a very high volume of contracts (914,000 ADV in Q1 2025 for FX rate derivatives). While not directly comparable to the US notional value without knowing the average contract size and price, it points to a very active market, likely driven by the volatility and hedging needs associated with the Brazilian Real.
4. Russia (MOEX): The estimated turnover (~$51 billion for the month) suggests a large market, likely dominated by USD/RUB and potentially EUR/RUB pairs, reflecting Russia's trade patterns and the currency's volatility. Note that this figure is an estimate based on the total derivatives market share.
5. Asia (Ex-India):
    - Japan (JPX): Shows significant monthly volume (over 2 million contracts), indicating active trading, likely involving the Yen against the USD and other majors.
    - Singapore (SGX): Also displays high monthly volume (3.79 million contracts), cementing its role as a major Asian FX hub.
    - Hong Kong (HKEX): Shows lower activity compared to Singapore and Japan in this specific dataset (24,010 ADV contracts), though it remains a key offshore Renminbi hub (much of which might be OTC rather than exchange-traded futures).
6. Europe (Eurex - Germany): The reported volume for FX derivatives on Eurex is notably low compared to its dominance in other asset classes (like interest rates). This suggests exchange-traded currency futures are not a primary focus for this exchange, with activity potentially concentrated elsewhere in Europe (like London's OTC market) or on other platforms.
7. South Africa (JSE): Data was unavailable in the provided search results, but the JSE is known to have an active ZAR currency derivatives market.

Potential Implications:

- High turnover generally indicates good market liquidity, making it easier and cheaper for participants to enter and exit positions.
- Active currency futures markets reflect significant international trade, investment flows, and the resulting need for businesses and investors to hedge currency risk.
- The volume in emerging market currencies highlights their increasing integration into the global financial system and the specific risks/opportunities they present.

In summary, the US (CME) dominates the exchange-traded currency futures landscape in value terms, but significant and vital markets exist in India, Brazil, Russia, and key Asian hubs like Japan and Singapore, catering to specific currency pairs and regional needs. However, careful attention must be paid to the differing metrics and reporting periods when comparing these markets.

India

- In July 2022, the monthly notional turnover of currency derivatives (futures + options) in India (National Stock Exchange of India + Bombay Stock Exchange + Metropolitan Stock Exchange of India) stood at ₹ 32,04,276 crore (i.e., about ₹32 lakh crore) compared to ₹26,95,814 crore in June 2022, according to SEBI India.
- For June 2022, the average daily turnover in the currency-derivatives segment was ~ ₹334 billion (₹33,400 crore) as per NSE India.
- Note: These figures cover currency derivatives broadly (futures + options) and not exclusively listed currency futures.
- More recently, regulatory changes (e.g., by the Reserve Bank of India) impacting hedging eligibility have caused volumes in exchange-traded rupee derivatives to drop dramatically.

Region	**Approximate turnover**
India (currency derivatives)	₹32 lakh crore in July 2022 (~₹320 trillion)
India (currency futures only)	₹334 billion/day (June 2022)
Global (FX market)	US$7.5 trillion/day (April 2022)
Global (exchange-traded FX derivatives)	US$150 billion/day (estimate)

Important caveats

- The Indian figures given combine futures and options in the currency-derivatives segment; separating out just the futures leg is harder to locate in public sources.
- Global figures are dominated by OTC FX; exchange−traded currency futures are a much smaller portion and less frequently broken out in published aggregate stats.
- Indian numbers reported by SEBI / exchanges are notional turnover in ₹ crore and typically present currency derivatives (futures + options) instrument-wise. Exchanges sometimes publish fiscal-year aggregates; SEBI's monthly/annexure tables give year totals per exchange.

India — exchange-traded currency derivatives (annual notional turnover, ₹ crore)

We have used SEBI's annexure tables (instrument-wise turnover for BSE / NSE / MSEI) and the exchanges' published stats. The two most reliable recent year totals we could extract directly from SEBI / exchange tables are below (I list each exchange and the combined total):

Fiscal year (April→Mar)	**NSE (₹ crore)**	**BSE (₹ crore)**	**MSEI (₹ crore)**	**Combined (₹ crore)**
2023-24	3,36,24,479	22,88,115	2,47,088	3,61,59,682
2024-25	15,00,784	39,376	54,103	15,94,263

Source: SEBI Annexure tables.

We can see a very large fall in exchange-traded currency-derivatives turnover between FY2023-24 (₹361.6 lakh crore ~ ₹36,15,9682 crore) and FY2024-25 (₹15.94 lakh crore). That collapse is documented in press coverage and SEBI notes and is primarily driven by the RBI's April 2024/2025 restrictions on who may trade rupee derivatives on exchanges (limiting to those with underlying hedging exposure), which pushed most client business back into the OTC forward market. Reuters summarised this change and its effect.

Global — FX market and exchange-traded FX derivatives (annual / daily averages)

Bank for International Settlements (BIS) Triennial snapshots are the canonical source for global FX turnover (OTC). Exchange-traded FX futures/options are much smaller than OTC FX; public aggregates for exchange-traded FX are not reported in the same detail globally.

Yearly (BIS snapshot / annual)	Global OTC FX turnover (daily avg)	Rough estimate exchange-traded FX derivatives (daily avg ≈ 2% of OTC)
2019 (BIS Apr 2019)	$6.6 trillion / day	$132 billion / day
2022 (BIS Apr 2022)	$7.5 trillion / day	$150 billion / day
2025 (BIS Apr 2025 snapshot published Oct/Nov 2025)	$9.6 trillion / day	$192 billion / day

Source: Bank for International Settlements (BIS)

Interpretation/explanation

- The OTC FX market is huge (trillions per day) and grew from ~US$6.6tn/day (2019) → ~US$7.5tn/day (2022) → ~US$9.6tn/day (2025 snapshot). Those are OTC totals (spot, forwards, swaps, options).
- Exchange-traded FX futures/options represent a small fraction of total FX turnover — my 2% figure is a rough, commonly used rule-of-thumb to show scale; regional markets (e.g., CME, ICE, NSE IFSC) will show their own exchange volumes. If you need precise exchange-traded FX futures turnover by exchange (CME, ICE, Eurex, NSE IFSC, SGX, etc.) I can extract exchange annual reports/statistics and present a per-exchange 5-year table.

→ VOLATILITY ANALYSIS:

An overview and why GARCH(1,1) was used.

We used a two-pronged approach to analyze the volatility of the USD/INR futures market. Firstly a static analysis was performed by finding the annualized standard deviations of futures[1] and spot[2] prices, along with the correlation (Pearson's correlation/ ρ) to determine the Optimal Hedge Ratio (OHR). Next, a dynamic analysis using Generalized Autoregressive Conditional Heteroskedasticity (GARCH(1,1)) model to meet the project's need for a volatility model (MV) by capturing the changing and lasting nature of market risk. We used 2,407 observations of settlement prices to calculate daily logarithmic returns, which we used for all of our analysis.

Descriptive statistics showed that the GARCH model was needed. The futures returns displayed substantial leptokurtosis (Kurtosis ≈ 3.70) and a highly significant Jarque-Bera statistic (p-value ≈ 0.000000), thereby formally rejecting the hypothesis of a normal return distribution. This showed that there were fat tails and volatility clustering, which are situations in which the simple historical standard deviation greatly underestimates the actual market risk. So, the GARCH(1,1) model was chosen as the best way to find out how time-varying conditional volatility (σ_t) works and to formally check for volatility persistence, which is what financial econometrics does.

Estimated results

1. Static Volatility and Risk Management

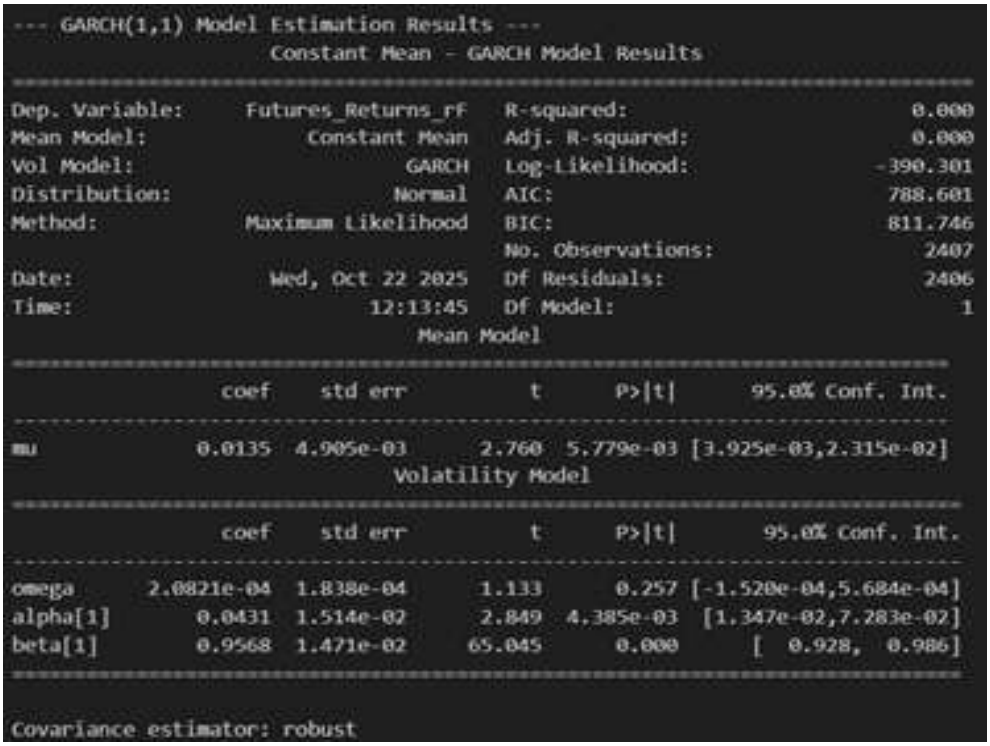
The initial static analysis quantifies the baseline risk and hedging potential of the USD/INR futures contract.

Metric	**Value**	**Interpretation**
Annualized Futures Volatility (σ_F)	**0.0496 (or 4.96%)**	The annual price risk (standard deviation) of the USD/INR Futures contract is 4.96%. This is the expected range of price fluctuation over a year, implying a relatively low-risk asset compared to equities (which often have volatilities of 15-25%).
Annualized Spot Volatility (σ_S)	**0.0510 (or 5.10%)**	The annual price risk of the USD/INR Spot exchange rate is 5.10%.
Comparison σ_S vs. σ_F	$\sigma_S > \sigma_F$	The Spot market is slightly more volatile than the Futures market. This is a common finding, suggesting that the Futures market may absorb risk efficiently or that the futures pricing mechanism smooths out some of the extreme daily swings.

Correlation (ρ)	**0.7688**	This is a strong positive correlation. It means that if the Spot price increases by 1%, the Futures price tends to increase by 0.7688%. The two markets move very closely together, which is essential for effective hedging.

The Optimal Hedge Ratio (OHR) was found to be 0.7901 based on these results. This means that a hedger needs to take a futures position worth $0.7901 for every $1 of uncovered spot exposure in order to keep portfolio variance as low as possible. The Hedging Effectiveness (HE), which is ρ^2, is 59.10%. This strong percentage shows that the USD/INR futures market is a very useful risk management tool for people who trade in it, as it can cut price variance by more than half.

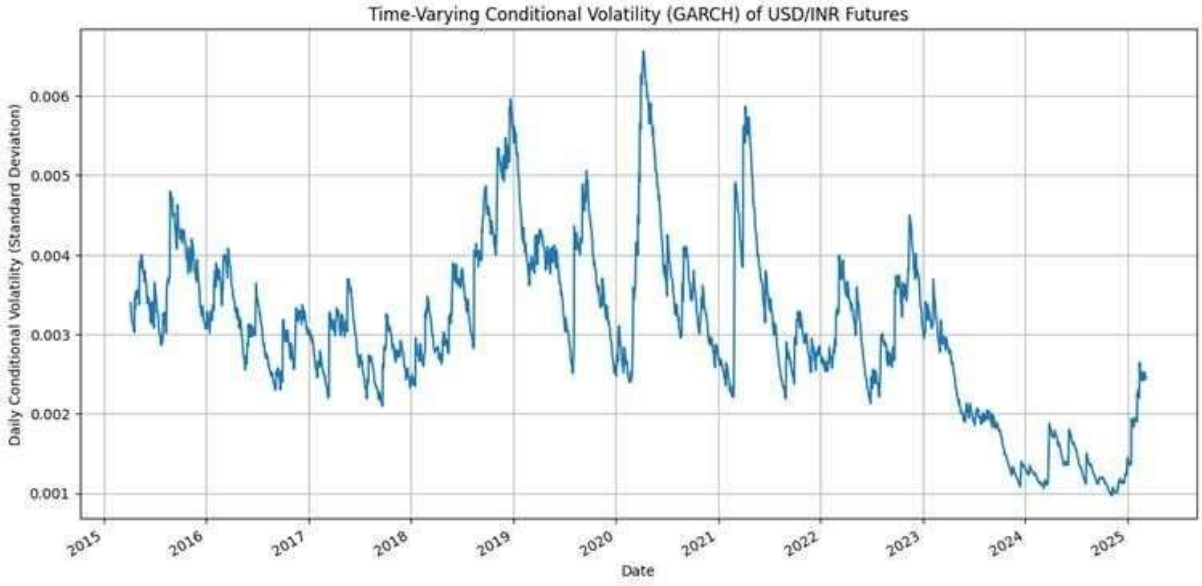
2. Dynamic Volatility Analysis (GARCH)



```
--- GARCH(1,1) Model Estimation Results ---
                  Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:     Futures_Returns_rf   R-squared:                       0.000
Mean Model:             Constant Mean   Adj. R-squared:                  0.000
Vol Model:                      GARCH   Log-Likelihood:               -390.301
Distribution:                  Normal   AIC:                           788.601
Method:            Maximum Likelihood   BIC:                           811.746
                                        No. Observations:                 2407
Date:                Wed, Oct 22 2025   Df Residuals:                     2406
Time:                        12:13:45   Df Model:                            1
                            Mean Model
==============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
------------------------------------------------------------------------------
mu             0.0135  4.905e-03      2.760  5.779e-03 [3.925e-03,2.315e-02]
                          Volatility Model
==============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
omega      2.0821e-04  1.838e-04      1.133      0.257 [-1.520e-04,5.684e-04]
alpha[1]       0.0431  1.514e-02      2.849  4.385e-03  [1.347e-02,7.283e-02]
beta[1]        0.9568  1.471e-02     65.045      0.000      [  0.928,  0.986]
==============================================================================

Covariance estimator: robust
```

The sum of the coefficients, ($\alpha_1 + \beta_1$), is close to one, which is typical of financial volatility that lasts a long time. This means that a risk shock that enters the USD/INR market doesn't go away quickly; instead, the increased volatility lasts for a long time. The large value of the α_1 coefficient shows that people in the market react very strongly to big mistakes (shocks), which directly leads to more volatility in the future. This high level of persistence is an important finding for risk managers because it means that simple, short-term hedging strategies might not work when the system is under stress.



The market's dynamic risk profile is best visualized by the GARCH-derived Conditional Volatility (σ_t) series. Market risk is extremely uneven, varying greatly around the average 4.96% level, according to the Conditional Volatility plot. The following exhibit the biggest spikes in volatility:

Early 2020: In response to the first worldwide shock of the COVID-19 pandemic and the subsequent flight to safety, which caused significant volatility in the exchange rate.

Early 2021: Linked to a period of global inflationary recovery and growing apprehension regarding interest rate policy changes by central banks (the Federal Reserve and the Reserve Bank of India). The plot emphasizes that although the market is generally stable, there are severe periods of extreme risk that necessitate regulatory or central bank intervention to restore stability because of the GARCH persistence (β_1). The conditional volatility has levelled off close to its lowest point in the sample as of late 2024/early 2025, indicating a decrease in immediate market risk.

The Volatility-Liquidity Nexus: A GARCH-Volume Synthesis

The analysis of both Conditional Volatility (σ_t) and Annual Trading Volume reveals a strong positive relationship between market risk and liquidity in the USD/INR futures market.

• Risk-Driven Participation: The Conditional Volatility graph, based on GARCH(1,1), highlights distinct spikes in volatility during early 2020 due to the COVID-19 impact and in 2021 because of concerns over global inflation and interest rate policies.

• Liquidity Peak: These periods of heightened risk align with the Peak Liquidity Phase from 2021 to 2022, as illustrated in the Annual Volume chart, where contract volumes surpassed 70 million.

• Contraction: Conversely, the observed stabilization of conditional volatility in late 2024 and early 2025 corresponds with the subsequent Liquidity Contraction Phase in the volume data.

This relationship provides significant insights into the market. Volatility influences both information flow and incentives. During times of high volatility, hedgers seek to mitigate risk more aggressively, while speculators identify greater profit opportunities. This interplay results in increased market volume and tighter spreads, which reflect enhanced liquidity. The market reaches its highest liquidity when the GARCH model indicates maximum daily risk.

Connection with theoretical underpinnings

The volatility analysis confirms the theoretical underpinnings of currency futures while providing crucial empirical nuance regarding market efficiency and risk.

The Cost of Carry Model and Arbitrage Pricing are upheld by the static analysis. The key theoretical connection needed to stop covered interest arbitrage opportunities, the mechanism that keeps the Interest Rate Parity (IRP) in equilibrium, is validated by the strong positive correlation (ρ =0.7688) between spot and futures returns. Additionally, as stated by Hedging Theory, currency futures serve as effective instruments for risk management, as demonstrated by the derived Optimal Hedge Ratio (OHR) of 0.7901, which yields a Hedging Effectiveness (HE) of 59.10%.

However, the dynamic GARCH (1,1) analysis qualifies the market's efficiency. The assumption of a smooth Efficient Market Hypothesis (EMH) where returns are normally distributed is contradicted by the severe leptokurtosis and rejection of the normal distribution (p-value ≈ 0.000000). A more nuanced perspective is offered by the GARCH model, whose significant α_1 coefficient (0.0431) validates the EMH by demonstrating that volatility responds instantly to new information6666. Importantly, the near-unity sum (0.9999) and the β_1 coefficient (0.9568) both show high persistence, which contradicts the simple Expectations Hypothesis and suggests that the market requires a risk premium for taking on non-decaying risk. This high persistence suggests that shocks cause prolonged periods of uncertainty, which should be a major focus for policy implications. This is demonstrated by the conditional volatility plot's long decay of volatility spikes (such as post-COVID 2020/2021).

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[1] https://in.investing.com/currencies/usd-inr-futures-historical-data

[2] https://in.investing.com/currencies/usd-inr-historical-data

VI. CASE STUDY: A STRUCTURAL AND QUANTITATIVE COMPARISON BETWEEN JAPANESE FUTURES MARKET AND INDIAN FUTURES MARKET

Based on the data extracted from the CME, the comparison between the Indian and Japanese currency futures markets reveals pronounced differences in both structural design and quantitative performance. The Japanese yen futures market shows much higher levels of trading activity and open interest compared to the Indian rupee futures market, showing its superior liquidity and market depth. For contracts dated December 2025, Japan's yen futures recorded a total trading volume of 197,891 contracts and an open interest of 304,985, while the Indian rupee futures market reported a total volume of 441 in contracts dated October 2025 and an open interest of just 1,248. This gap highlights the broader and more active participation of institutional investors, multinational corporations, and global speculators in Japan's futures market, supported by the yen's position as a major reserve and trade currency worldwide. The smaller day-to-day price movements in Japan's market, such as a change of −0.02% in November and +0.12% in December 2025, indicate lower short-term volatility and a stable, mature trading environment. In sharp contrast, India's market saw a consistent price change of +0.41% across consecutive months, suggesting higher short-term volatility—an attribute often associated with emerging markets that are more exposed to fluctuations in capital flows and macroeconomic shifts.

From a structural point of view, Japan's currency futures are designed for a global investor base. Each contract represents 12,500,000 Japanese yen, quoted in U.S. dollars and cents per yen, and is settled through physical delivery. This deliverable structure makes the contracts more effective for corporations and investors managing real trade or investment exposures, while the availability of contracts for up to eight consecutive quarters provides flexibility for long-term hedging. In comparison, the Indian rupee futures market follows a financial settlement model, where contracts are cash-settled in rupees rather than through the physical exchange of currency. Each contract covers 5,000,000 INR, quoted in U.S. cents per 100 INR, and is listed monthly for twelve consecutive months with four additional quarterly contracts. This structure is better suited to short-term hedging and speculative trading rather than long-duration exposure management. The difference in settlement type—deliverable in Japan and financial in India—marks a fundamental difference in market design and participant behavior: while Japan's market is closely linked to actual trade and investment flows, India's market operates more as a regulated financial instrument geared toward domestic hedging and price discovery.

Overall, the metrics from the CME data show that the Japanese currency futures market is more liquid, deeper, and less volatile, supported by higher turnover, extensive global participation, and long-standing regulatory confidence. The Indian market, although smaller, continues to expand and diversify, showing increasing activity but also greater price variability—signs of an emerging system still gaining depth and maturity. These contrasts also reflect regulatory differences: Japan's futures operate within a highly developed and internationally connected framework under the oversight of the Financial Services Agency (FSA) and the Bank of Japan, while India's currency derivatives market is jointly regulated by SEBI and the RBI, with stricter participation norms and an emphasis on market stability. In essence, both markets serve as important avenues for managing currency risk, but Japan's futures market embodies the characteristics of a well-established global financial hub, whereas India's market represents a rapidly evolving ecosystem steadily moving toward greater integration, efficiency, and resilience.

Source: Search - CME Group

VII. CHALLENGES:

1. Market Risk (Volatility)

In global markets, currency futures on major pairs like USD/EUR or USD/JPY tend to experience sharp swings whenever central banks adjust interest rates or when geopolitical tensions rise. This volatility is both a challenge and an opportunity, attracting speculators but making risk management tricky for corporations. In India, however, volatility in INR futures is somewhat muted because the Reserve Bank of India often intervenes to stabilize the rupee. While this helps prevent sudden shocks, it also makes the futures market less reflective of true demand and supply conditions.

2. Basis Risk

Globally, the basis risk gap between spot and futures prices is usually small because arbitrage opportunities are quickly exploited in highly efficient markets. Traders can move between futures and spot markets easily, keeping prices aligned. In India, however, basis risk tends to be higher. Regulatory restrictions and RBI's influence on currency markets mean arbitrage is less effective, so the futures and spot prices may not move in sync, making hedges less reliable.

3. Liquidity Risk

Internationally, liquidity is rarely an issue for major currency futures, as high trading volumes and large market participation ensure tight spreads and smooth execution of trades. Only exotic or thinly traded currencies face liquidity problems. In contrast, India's futures market is liquid mainly for USD/INR contracts, while pairs like EUR/INR, GBP/INR, and JPY/INR see far lower activity. Limited participation from foreign players also makes liquidity shallower compared to global exchanges, which can push up costs for traders and hedgers.

4. Margin and Leverage

One of the defining features of global currency futures markets is the ability to trade with high leverage and relatively low margin requirements. This allows traders to control large positions with minimal capital, but it also exposes them to sudden margin calls during volatile periods, amplifying losses. India takes a much more conservative approach. SEBI and the RBI impose stricter margin rules and keep leverage relatively low. While this protects the financial system from instability, it reduces the appeal of futures for speculators and aggressive traders.

5. Standardization Issues

Futures are standardized everywhere, but global exchanges often offer a wide range of contract sizes, including mini and micro contracts, so that both retail traders and large institutions can participate. This gives more flexibility to match exposures. India, however, offers little flexibility in its contracts. For example, USD/INR contracts are fixed at $1,000 with set monthly expiries. This rigid structure doesn't suit corporates with unique hedging needs, pushing them to use over-the-counter (OTC) forwards instead of exchange-traded futures.

6. Transaction Costs

Globally, transaction costs in futures markets are relatively low. Intense competition among exchanges, high trading volumes, and advanced electronic platforms ensure that spreads are narrow and fees remain minimal. India, on the other hand, has relatively higher costs. Along with brokerage and exchange fees, traders also pay the Securities

Transaction Tax (STT). For less liquid contracts like EUR/INR or GBP/INR, wider bid-ask spreads add to the overall cost, reducing the efficiency of trading.

7. Regulatory Environment

Global futures markets are relatively open and liberal, allowing participation from hedge funds, corporates, banks, and speculators alike. Regulators such as the CFTC in the US or the FCA in the UK primarily ensure transparency and fair practices without overly restricting access. In India, the environment is much more controlled. Both SEBI and the RBI impose strict rules on participation, with limited access for foreign investors and restrictions on how contracts can be used. Since the rupee is not fully convertible, the Indian currency futures market is less integrated with the global system.

8. Counterparty and Systemic Risks

In global markets, clearinghouses like CME Clearing guarantee futures trades, which reduces counterparty risk significantly. However, systemic risks remain during global crises when clearing systems themselves may come under stress. In India, clearing corporations under exchanges such as NSE and BSE also guarantee trades, keeping counterparty risk minimal. Additionally, because of stricter regulations and limited foreign participation, India's futures market is relatively insulated from large-scale global shocks, though at the cost of reduced market depth.

9. Hedging Efficiency

For major global currencies, futures provide an efficient way to hedge exposure, thanks to deep liquidity and contracts with maturities that often align with business needs. Problems only arise with exotic currencies that lack trading volumes. In India, hedging efficiency is weaker because of the limited choice of contracts, thin liquidity outside USD/INR, and rigid standardization. Large corporates usually prefer OTC forwards with banks because they can tailor the contract exactly to their exposure, while smaller businesses and retail traders often use exchange-traded futures.

10. Economic and Political Risks

Globally, futures prices react to economic data releases, central bank decisions, and geopolitical developments. The markets are free-floating, and prices adjust quickly to reflect global events. In India, however, political and regulatory factors carry more weight. RBI interventions in the currency market, sudden changes in capital flow regulations, or policy decisions on trade can strongly influence INR futures. Since the rupee is not fully convertible, Indian currency futures do not always behave like those in open global markets, making them less predictable.

VIII. POLICY AND REGULATORY IMPLICATIONS:

1. Systemic Risk Management and Volatility Persistence

The GARCH (1,1) assessment uncovered a pivotal aspect of the USD/INR futures market, which is the extreme persistence of volatility ($\alpha 1 + \beta 1$ is near-unity, specifically ≈ 0.9999). This signifies that a risk shock entering the market would not be reversed quickly; on the contrary, the market would experience increased volatility for a long time.

Implication for the Regulators (RBI/SEBI):

Volatility persistence of high level ($\beta 1$ coefficient is 0.9568) means that during the stressful market periods, the hedging strategies have to be more sophisticated and longer-band. The policy has to consider the prolonged uncertainty.

Policy Action:

It may be the case that during the high-stress periods, such as the COVID-19 shock in early 2020, the regulators will need to take proactive measures to stabilize the market, perhaps through the following means:

· Adjustments to margin requirements on a temporary basis to control the activities of speculators that are excessive and thereby reduce volatility.

· Circuit breakers or enhanced surveillance to manage the long decay of volatility spikes could be implemented.

2. Impact of Regulatory Changes on Liquidity

Changes in regulations that are recent such as the restrictions imposed by the Reserve Bank of India (RBI) in April 2024/2025 regarding who is allowed to trade with rupee derivatives on exchanges (the train of thought being those with underlying hedging exposure) resulted in the exchange-traded currency-derivatives turnover taking a "very large fall".

Observed Impact:

The annual notional turnover in Indian exchange-traded currency derivatives dropped catastrophically from ₹3,61,59,682 crore in FY2023-24 to ₹15,94,263 crore in FY2024-25. Most client transactions were thus forced back into the Over-The-Counter (OTC) forward market.

Implication:

The issue of market participation restrictions imposed by policymakers will be examined, as such measures, though often aimed at curbing speculation, can inadvertently result in significant liquidity shortages in exchange-traded markets. This reduction in liquidity may, in turn, lead to higher transaction costs and wider bid-ask spreads.

Policy Review:

Regulators need to keep on doing the trade-off analysis in between speculation control and exchange market liquidity and transparency, for the reason that active futures markets are indispensable for risk management.

Global and Market Structure Implications

3. Role of Exchange-Traded vs. OTC Markets

The worldwide trend of forex derivatives traded through exchanges is a "very small segment" compared to the gigantic OTC forex market. The global OTC forex market is projected to be $9.6 trillion per day in 2025, whereas the exchange-traded forex derivatives are a rough estimate of just $192 billion per day (approximately 2% of the OTC total).

Implication:

The CME Group in the US controls the market of exchange-traded FX derivatives with $101 billion ADNV. In most parts of the world, including Europe (Eurex), the most important currency risk management is done in the non-transparent OTC market (for instance, London's OTC market).

Policy Suggestion:

Policies that aim to promote stability and transparency could consider either the transition of suitable OTC products to exchanges or the imposition of stricter reporting requirements on OTC transactions to allow better monitoring of the concentration of systemic risk.

4. Liquidity as an Essential Component of Risk Management

A high turnover in the market is an indicator of good liquidity which allows the participants to hedge their positions at lower costs and with less difficulty. The study of the USD/INR pair also confirmed that volume serves as a reliable indicator of liquidity.

Consequence:

The correlation between USD/INR spot and futures is very strong with a coefficient of 0.7688 and the calculated Optimal Hedge Ratio of 0.7901 (leading to a Hedging Effectiveness (HE) of 59.10%) is thereby proving that currency futures are very effective risk management tools.

Regulatory Support:

Regulators must see to it that the liquid markets in India, Brazil (B3), and Asia (Japan/Singapore) which are for specific currency pairs (e.g., USD/INR, BRL) and regional hedging demands continue to thrive. Furthermore, active currency futures markets are necessary to cope with the risk arising from international trade and investment flows.

Theoretical and Modeling Implications

5. Need for Advanced Volatility Modeling in Regulation

The distribution of market returns was found to be non-normal, since it had a leptokurtic shape. Thus, the simple historical standard deviation greatly underestimates the actual market risk.

Implication for Models:

Traditional risk models that are based on static volatility (like simple VaR models) will give wrong results during stress periods.

Policy Action:

Risk management frameworks of regulations and exchanges (e.g. for margin calculation) should not be limited to simple historical volatility models anymore but rather employ dynamic ones such as GARCH (1,1). The Conditional Volatility plot gives a picture that market risk is "extremely uneven".

6. The Simple Expectations Hypothesis is Challenged

The GARCH model's high persistence (near-unity sum of $\alpha 1 + \beta 1$) directly opposes the simple Expectations Hypothesis.

Result: It may imply that the market still requires a risk premium for accepting non-decayed risk.

Regulatory Attention: It is advisable for policy makers to investigate how risk premiums are established and subsequently valued in the futures market, since this understanding is imperative not only for recognizing the actual price of hedging but also for creating sound protection policies for investors.

IX. CONCLUSION

This project has essentially tracked the entire story of currency futures, starting from when they were what the market desperately needed after the Bretton Woods system fell apart, to their current role as the go-to tools for managing financial risk. Watching them evolve from risky, one-off forward deals to transparent, exchange-traded contracts shows how the market solved the massive uncertainty that floating exchange rates created.

Now, while the textbooks paint a picture of a smooth, logical market—all based on Interest Rate Parity and Cost of Carry models—our own number-crunching reveals a much messier and more complicated truth. The data clearly isn't "normal" (the leptokurtosis is a dead giveaway), which confirms that the simple, old-school volatility models just don't come close to capturing the real-world risk.

The biggest takeaway from the data is how stubbornly persistent volatility is in the USD/INR market. Our GARCH model showed this clearly: the $\alpha 1$ and $\beta 1$ coefficients add up to almost 1 (0.9999), meaning shocks don't just disappear. They dig in and cause long hangovers of uncertainty. You can see it plain as day in the volatility spikes from 2020-2021. It proves that liquidity isn't a given; it's something that comes and goes based on that very volatility.

But even with all that, these tools are incredibly useful. The analysis proved they do the job: with an Optimal Hedge Ratio of 0.7901 and a solid 59.10% Hedging Effectiveness, they are clearly effective at cutting down portfolio risk.

However, the Indian market is its own animal. Our comparison with Japan made it obvious that India’s market is built differently, favoring cash settlement over physical delivery. It's still growing up, and it's a bumpier ride than Japan's deep, liquid, global hub. Piled on top of this are India's specific challenges, like poor liquidity outside the main USD/INR pair, higher basis risk, and the RBI constantly stepping in, which both helps and hurts.

Ultimately, this entire project boils down to one critical policy paradox. India's 2008 introduction of futures was a big win for transparency. But the RBI's recent (April 2024/2025) crackdown on who can participate has absolutely tanked the trading volumes. This move, meant to cut down speculation, has accidentally sent everyone running back to the opaque, shadowy OTC forward market. It completely undermined the very stability the exchanges were supposed to build. The future of India's futures market now all comes down to whether the regulator can walk this tightrope between taming speculation and preserving the liquidity that everyone needs to manage their risk.

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